Problem Set (MABE)

Cost functions

1. Suppose a firm must pay an annual tax, which is a fixed sum, independent of whether it produces any output.
    a. How does this tax affect the firm's fixed, marginal and average costs?
    b. Now suppose the firm is charged a tax that is proportional to the number of items it produces. Again, how does this tax affect the firm's fixed, marginal, and average costs?
2. The short-run cost function of a company is given by the equation TC = 200 + 55q, where TC is the total cost and q is the total quantity of output, both measured in thousands.
    a. What is the company's fixed cost?
    b. If the company produced 100,000 units of goods, what would be its average variable cost?
    c. What would be its marginal cost of production?
    d. What would be its average fixed cost?
    e. Suppose the company borrows money and expands its factory. Its fixed cost rises by $ 50,000, but its variable cost falls to $45,000 per 1,000 units. The cost of interest (i) also enters into the equation. Each 1-point increase in the interest rate raises costs by $3,000. Write the new cost equation.
3. The firm's total cost function is $TC = 400 + 2Q + 0.01Q^2$. Thus its minimum efficient scale of production (MES) is at Q = ___.

Perfect Competition

1. New England textiles, Inc. is a medium-sized manufacturer of blue denim that sells in a perfectly competitive market. The total cost function for this product is described by $TC = 250 + Q + 0.25Q^2$. where Q is square yards of blue denim produced per month.
    a. Derive the firm's supply function, expressing quantity as a function of price.
    b. If there are 500 identical firms in the industry, derive the market supply function.

2. The demand for a type of carpet known as KP-7 has been estimated as:

    P = 40 -0.25Q where P is price and Q is quantity.

    The market supply is expressed as:

    P = 5 +0.05 Q.

    A typical firm in this market has a total cost function given as:

    TC = 100 – 20q + 2 q^2

    a. Determine the equilibrium market output and price.
    b. Determine the output for a typical firm.
    c. Determine the profit (or loss) earned by the typical firm.
    d. Derive the firm's short-run supply function.

3. The handmade snuffbox industry is composed of 100 identical firms, each having short-run total costs given by $STC = 0.5q^2 + 10q + 5$, where q is the output of snuffboxes per day.
    a. What is the short-run supply curve for each snuffbox maker?
    b. What is the short-run supply curve for the market as a whole?
    c. Suppose the demand for total snuffbox production is given by $Q = 1100 - 50P$. What is the equilibrium in this marketplace? What is each firm's total short-run profit?
    d. Graph the market equilibrium and compute total producer surplus in this case.
    e. Suppose now that the government imposed a $3 tax on snuffboxes. How would this tax change the market equilibrium?
    f. Calculate the total loss of producer surplus as a result of taxation of snuffboxes. Show that this loss equals the change in total short-run profits in the snuffbox industry. Why don't fixed costs enter into this computation of the change in short-run producer surplus?

4. Consider a competitive market in which the market demand for the product is expressed as $P = 75 - 1.5Q$, and the supply of the product is expressed as $P = 25 + 0.50Q$. Price, P, is in dollars per unit sold, and Q represents rate of production and sales in hundreds of units per day. The typical firm in this market has a marginal cost of $MC = 2.5 + 10q$
    a. Determine the equilibrium market price and rate of sales.
    b. Determine the rate of sales of the typical firm, given your answer to 3.1.
    c. If the market demand were to increase to $P = 100 - 1.5Q$, what would the new price and rate of sales in the market be? What would the new rate of sales for the typical firm be?

Monopoly and Monopsony

1. A monopolist faces a market demand curve given by Q = 70 – P. The monopolist's marginal revenue is given by MR = 70 -2Q. If the monopolist can produce at constant average and marginal costs of AC = MC = 6 (assuming that there is no fixed cost).
    a. What output level will the monopolist choose in order to maximize profits?
    b. What is the price at this output level?
    c. What are the monopolist's profits?

2. Assume instead that the monopolist has a cost structure where total costs are described by TC = 0.25 Q^2 – 5Q + 300. With the monopolist facing the same market demand and marginal revenue (as in question 1), what price-quantity combination will be chosen now to maximize profits? What will profit be?

3. A single firm monopolizes the entire market for Batman masks and can produce at TC = 10Q. Originally, the firm faces a market demand curve given by Q = 60 – P.
    a. Calculate the profit-maximizing price-quantity combination for the firm. What are the firm's profits?
    b. Now assume that the market demand curve becomes steeper and is given by Q = 45 – 0.5P. What is the firm's profit-maximizing price-quantity combination now? What are the firm's profits?
    c. Instead of the assumptions in part b, assume that the market demand curve becomes flatter and is given by Q = 100 – 2P. What is the firm's profit-maximizing price-quantity combination now? What are the firm's profits?
    d. Graph the three different situations of part a, part b, and part c. Using your results, explain why there is no meaningful "supply curve" for this firm's mask monopoly.

4. Suppose a monopoly can produce any level of output it wishes at a constant marginal (and average) cost of $5 per unit. Assume the monopoly sells its goods in two different markets separated by some distance. The demand curve in the first market is given by $q_1 = 55 - p_1$ and the demand curve in the second market is given by $q_2 = 70 - 2p_2$
    a. If the monopolist can maintain the separation between the two markets, what level of output should be produced in each market, and what price will prevail in each market? What are total profits in this situation? Also compare the price elasticities of demand at the equilibrium.
    b. How would your answer change if the firm was forced to follow a single-price policy?

5. A firm has 2 factories, for which costs are given by $TC_1 = q_1^2 - 5q_1 \; and \; TC_2 = 0.5q_2^2 - 5q_2$. The firm faces the following demand curve $Q = 100 - P$. How much total output will the firm choose to produce and how will it distribute this production between its two factories in order to maximize profits? Find the firm's profits.

6. A monopolist can produce at a constant average and marginal cost of *SAVC* = *SMC* = 5 with a fixed cost of 100. The market demand curve is $Q = 35 - p$. What are profit-maximizing price and quantity?

7. The demand for T-Galaxy's product is: Q = 10 - 0.4P. T-Galaxy's marginal costs are MC = 3 + 6Q.
    a. Determine T-Galaxy's profit maximizing price.
    b. Calculate the deadweight loss from monopoly power.

c. If the monopoly decides to adopt a perfect price discrimination strategy, find the monopolist's profits. Is there any deadweight? Explain your answer.

8. The employment in a small village by one large factory can be characterized as a monopsony. Suppose the demand for labor function is $L^D = 500 - W$, where W is the wage, and L is the number of workers employed. The labor supply function is given by $L^S = 2W - 100$.
   a. How many workers will the factory hire? What wage will it pay?
   b. If the workers form a labor union, find the wage rate that the union would propose to the factory owner.

Oligopoly and Monopolistic Competition

1. Assume that there are only two firms (A and B) in the considered market. Let the demand and total cost functions be given by

$$P = 26 - 2Q$$

$$MC_A = AC_A = 5 \text{ and } MC_B = AC_B = 5$$

where the subscripts denote firm A and firm B

1.1 What are the Cournot Duopoly equilibrium price and quantity
1.2 What are the Stackelberg equilibrium price and quantity (Let A be a leader and B be a follower)
1.3 What are the cartel equilibrium price and quantity
1.4 Compare profits among the three equilibria and explain.

2. Assume that there are only two firms (A and B) in the market. Let the demand and total cost functions be given by

$$P = 100 - 0.5(Q_A + Q_B)$$

$$TC_A = 5Q_A \text{ and } TC_B = 0.5Q_B^2$$

Where the subscripts denote firm A and firm B

2.1 What are the Cournot Duopoly equilibrium price and quantity
2.2 What are the Stakelberg equilibrium price and quantity (let A be a leader)
2.3 What are the cartel equilibrium price and quantity
2.4 Compare profits among the three equilibria and explain.

3. Suppose that the total market demand for crude oil is given by $Q_D = -2000P + 70000$ where Q is the quantity of oil in barrels per year and P is the dollar price per barrel. Suppose

also that there are 1,000 identical small producers of crude oil, each with marginal costs given by $MC = q + 5$ where q is the output of the typical firm.

3.1 Assuming that each small oil producer acts as a price taker, calculate the typical firm's supply curve (q=…), the market supply curve (Q_S = …), and the market equilibrium price and quantity.

3.2 Suppose a practically infinite supply of crude oil is discovered in New Jersey by a would-be price leader and that this oil can be produced at a constant average and marginal cost of AC=MC=$15 per barrel. Assume also that the supply behavior of the competitive fringe described in (3.1) is not changed by this discovery. Calculate the demand curve facing the price leader.

3.3 How much should the price leader produce in order to maximize profits? What price and quantity will now prevail in the market?

4. Firm A and Firm B both produce differentiated products. Let firm A face demand curve $q_A = 100 - P_A + 0.5P_B$ and firm B face demand curve $q_B = 50 - P_B + P_A$. Products A and B both have constant marginal cost of production of 10 per unit (and no fixed cost). Each firm acts as a Bertrand competitor. What are the Bertrand Equilibrium prices in this market?

5. Firm 1 can produce at a constant average and marginal cost of *SAVC* = *SMC* = 5 with a fixed cost of 100. The market demand curve is $Q = 35 - P$.

5.1 Assume a second firm enters the market. Let q_1 be the output of firm 1 and q_2 the output of firm 2. Assuming firm 2 has the same costs as firm 1 and firm 2 behaves as a follower and firm 1 behaves as a leader, calculate the Stackelberg equilibrium output of each firm.

5.2 However, if both firms choose the Bertrand price competition, what will be the market outcome?

6. Assume there are <u>three</u> identical firms in the industry. The market demand is $P = 10 - Q$, where $Q = q_1 + q_2 + q_3$ and each firm has constant marginal (and average) cost of $ 4 per unit.

6.1 What are the Bertrand price and quantity

6.2 Compute the Cournot equilibrium (market price, each firm's output)

7. Consider a Bertrand model with product differentiation in which the demand and cost functions are $q_1 = 88 - 4p_1 + 2p_2,\ TC_1 = 10q_1$ and

$$q_2 = 56 + 2p_1 - 4p_2,\ TC_2 = 8q_2$$

for Firm 1 and Firm 2 respectively. Determine equilibrium price, quantity and profits for each firm.

8. The kinked demand curve is $Q = 360 - 40P$ over the range from 0 to 40 units of output. When output exceeds 40, $Q = 120 - 10P$. The firm's total cost function is $\mathrm{TC} = 10 + \mathrm{Q} + 0.0625Q^2$.

    8.1 Graph the firm's demand

    8.2 Calculate price and output at the profit maximizing level activity?

True, False or Uncertain and explain why. All points will be awarded based on the quality of your explanation.

1. A producer will determine whether to operate or not by considering her short-run average variable costs, her profit level by considering her short-run average total cost, and her optimal level of output by considering her short-run marginal cost.
2. A monopoly has no competition, so it can charge whatever price it wishes and it always enhances profit by increasing its price.
3. It costs more to increase output in the short-run than in the long-run.
4. A perfect price discrimination increases economic efficiency as compared to simple monopoly.
5. The market structure of the local pizza industry is best characterized by monopolistic competition. Mama Pizza is one of the producers in the local market. The demand for Mama Pizza is: $q_m = 225 - 10\,p_m$. Mama's cost function is: $TC_m = 0.15q_m^2$. Thus, Mama's profit maximizing level of output and price is at the long-run equilibrium.
6. If the fringe supply curve shifts leftward in the dominant firm model, then the resulting market equilibrium price is higher and the dominant firm's quantity increases.
7. Monopolist will never choose to operate at a point where demand is inelastic.
8. There is no deadweight loss when the monopolist adopts the first-degree price discrimination.